SPEAK-ALIGN: AI-Powered Call Script Compliance and Quality Assurance System

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1. Introduction

Background of the Study

Business Process Outsourcing (BPO) call centers, which handle millions of calls every day across a variety of industries, are vital touchpoints between businesses and their clients. Traditionally, manual monitoring has been used in call centers for quality assurance. Supervisors randomly select recorded calls to assess agent performance in relation to predetermined scripts and protocols. There are large gaps in quality monitoring and agent development opportunities due to this manual approach's time commitment, subjectivity, and limited coverage of the overall call volume.

Current Issues

Call centers current manual quality assurance process has a number of problems, such as limited call monitoring coverage (usually 2% to 5% of all calls), delayed agent feedback (often weeks after the call), inconsistent evaluation because of human bias, high operational costs for quality assurance teams, and the inability to detect script compliance issues in real-time. Additionally, because manual evaluation is subjective, agents who feel their assessments are unfair may argue and become demotivated.

Problem Statement

An objective, scalable, and automated system to check if agents are correctly and sequentially following call scripts is absent from BPO call centers. Management cannot maintain service quality standards, guarantee regulatory compliance, find training gaps, or give agents timely, helpful feedback without thorough monitoring. This leads to uneven client experiences, possible infractions of the law, and lost chances to enhance agent performance.

Proposed Solution

In order to confirm script compliance and offer useful feedback, this project suggests SPEAK-ALIGN, an AI-powered automated quality assurance system that examines 100% of call recordings. The system makes use of sophisticated speech-to-text technology for precise transcription, speaker diarization to distinguish between agent and customer voices, and large language models to intelligently compare agent transcriptions to standard scripts. Without the need for intricate dashboards or timestamp analysis, the system detects deviations, assesses script sequence adherence, and provides concise, helpful feedback for agent improvement

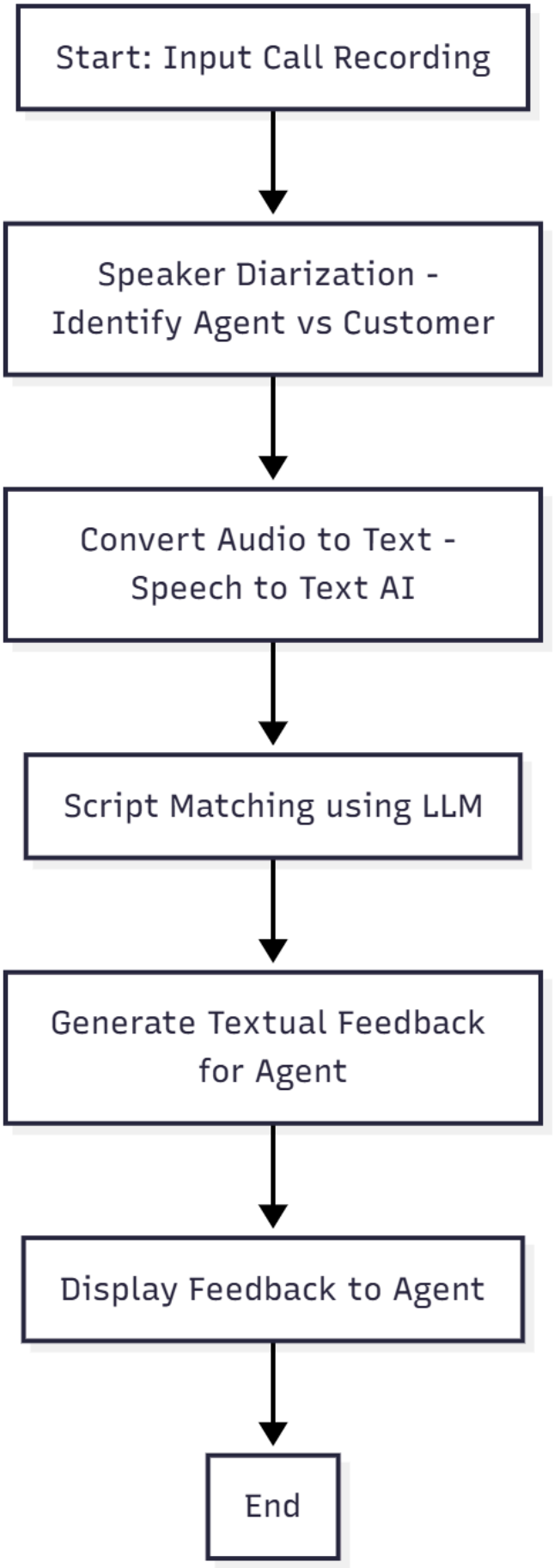
2. Existing System/Related Works

Table 1: Summary of Existing Systems/Work/Applications

Article No/App/System	**Year**	**Features**	**Limitations**
CallMiner Eureka	2023	Speech analytics, sentiment analysis, automated scoring	Scoring rules are not visible or editable, focuses more on keyword presence and predefined categories, CallMiner offers category-based insights but limited coaching detail
NICE Enlighten AI	2023	AI-driven quality management, interaction analytics	used for behavioural insights (agent behaviour, sentiment, intent) rather than strictly checking if an agent followed a script.
Observe.AI	2023	Automated QA scoring, agent coaching	Their scoring algorithms, Script comparison rules, Feedback style cannot be changed, they are rigid

3. Proposed System

SPEAK-ALIGN is an automated quality assurance system tailored for BPO call centers that ensures script compliance and improves agent performance. The system processes uploaded call recordings through a pipeline that includes speaker diarization, distinguishing between agent and customer voices; speech-to-text using advanced AI models; and intelligent script comparison with large language models. This AI analyzes if the agent followed the prescribed script correctly and in order, and it identifies specific deviations. Based on this analysis, the system generates simple actionable text-based feedback highlighting mistakes and suggesting improvements sans complex visualizations or timestamp references.


Start: Input Call Recording
Speaker Diarization - Identify Agent vs Customer
Convert Audio to Text - Speech to Text AI
Script Matching using LLM
Generate Textual Feedback for Agent
Display Feedback to Agent
End

3.1. Scope

Target speakers for SPEAK-ALIGN include:

- Automated processing of call recordings in common audio formats: WAV and MP3
- Speaker diarization to separate agent speech from customer speech
- Highly accurate speech-to-text transcription of agent conversations
- Comparison of agent transcriptions against pre-defined call center scripts
- Detection of script deviations, omissions, and sequencing violations
- Generation of text-based feedback reports for individual calls
- Support for standard BPO scripts, including greetings, verification, issue resolution, closing protocols
- Processing of calls in English language
- Batch processing capability for multiple recordings
- Export of feedback reports in text format for management review

Out of Scope:

- Real-time call monitoring during live calls
- Customer sentiment analysis
- Dashboard or graphical user interface
- Support of several languages other than English

3.2. Proposed System Modules

3.2.1. Audio Upload Module

Function 1: Audio File Upload It allows the user to upload recordings of calls in supported formats: WAV, MP3. The system checks the correctness of file format, size constraints, and audio quality in order to guarantee correct processing.

3.2.2. Speaker Diarization Module

Function 1: Voice Separation This module utilizes the speaker diarization algorithms to automatically identify and separate the agent's voice from the customer's voice within the call recording.

Function 2: Agent Audio Extraction Once speakers are identified, the system extracts only the agent's audio segments for further transcription and analysis, filtering out the speech of customers and background noise.

3.2.3. Speech-to-Text Conversion Module

Function 1: Audio Transcription Equipped with state-of-the-art speech recognition technology, this module converts the extracted agent audio into highly accurate text transcription, capturing the exact words spoken by the agent during the call.

3.2.4. Script Management Module

Function 1: Script Upload and Storage User can add and save standard call center scripts that agents are required to stick to

3.2.5. AI-Powered Script Comparison Module

Function 1: Script Compliance Analysis This core module sends the standard script and the agent transcription to a large language model with specific prompts so it can analyze whether the agent follows the script accurately or not.

Function 2: Sequence Verification This includes the AI model evaluating if the agent maintained the correct order of elements in the script: greeting before verification, issue resolution before closing, etc.

Function 3: Deviation Detection Specific deviations are identified by the system where the agent either omitted elements of the script that should have been used, added unauthorized statements, or used incorrect phrasing.

3.2.6. Feedback Generation Module

Function 1: Mistake Identification Report Based on the AI's analysis, this module generates a clear text-based report listing all identified mistakes, deviations, and script violations.

Function 2: Improvement Suggestions The system provides actionable feedback and suggestions on how the agent can improve their script adherence, including specific phrases they should have used or sequences they should have followed.

Function 3: Report Export Generated feedback reports can be exported as plain text files for distribution to agents, supervisors, or training teams.

3.2.7. Frontend Module

Function: Provides a lightweight, user-friendly interface for QA teams and supervisors to interact with the system using **Streamlit**. Key features:

1. File upload widget for audio files
2. Job submission button with progress indicators (basic)
3. Call feedback display: shows agent transcription and AI-generated feedback

3.3. Advantages of the Proposed System

- **100% Call Coverage:** SPEAK-ALIGN can analyze every single call recording. This will ensure comprehensive quality monitoring, whereas manual QA can only sample 2% to 5% of the calls.
- **Objective and Consistent Evaluation:** The evaluation criteria for calls are standardized using AI, leaving no room for human judgment and bias in assessing the quality.

- **Cost-Effective Solution:** Long-term savings in operational expenditures through minimising the need for large QA teams which manually monitor calls.
- **Scalability:** Suitable for expanding BPO operations because it can manage growing call volumes with ease and without necessarily increasing the number of resources.
- **Specific Actionable Feedback:** Contrasted with general subjective comments, this level of feedback provides specific, useful criticism regarding certain script deviations.
- **Identification of Training Gap:** Helps in pinpointing common mistakes committed by agents, guides the development of focused training curriculum and the development of better scripts.
- **Compliance Assurance**: Ensures regulatory and company policy compliance through verification that required disclosures and procedures are followed on every call.
- **Ease of implementation:** There's no complex dashboard or integrations that are required, making deployment and use very simple without extensive technical overhead.
- **Agent Development Tool:** provides continuous learning for agents to self-improve by understanding what they have done wrong and what needs to be improved.

3.4. Limitations/Disadvantages

- **Audio Quality Dependency:** System accuracy is dependent on clear audio recordings poor quality audio with excessive noise or multiple speakers may reduce diarization and transcription accuracy.
- **Language Limitation:** Currently supports only English language; multilingual call centers would require additional language models and training.
- **Limited Customer Interaction Analysis:** Focuses only on agent script compliance without analyzing customer sentiment, satisfaction, or interaction quality.
- **Computational Resources:** Processing large volumes of audio files requires adequate computational infrastructure for speaker diarization and transcription tasks.

3.5. Applications

- **BPO Call Centers**: Primary application for outsourcing companies handling customer service, technical support, and sales calls across various industries.
- **Telemarketing Operations**: Ensuring sales agents follow legally compliant scripts and disclosure requirements during outbound calling campaigns.
- **Customer Support Centers**: Monitoring technical support agents to ensure they follow troubleshooting protocols and provide accurate information.
- **Financial Services Call Centers**: Verifying compliance with regulatory disclosure requirements in banking, insurance, and investment services calls.

- **Healthcare Call Centers**: Ensuring HIPAA compliance and proper patient information handling protocols in medical appointment scheduling and telehealth services.
- **E-commerce Customer Service**: Monitoring return policies, refund procedures, and customer interaction quality for online retail support teams.
- **Training and Development Programs**: Using system feedback as training material for new agent onboarding and continuous skill development.
- **Quality Assurance Departments**: Augmenting QA team capabilities by automating initial script compliance checks, allowing human reviewers to focus on complex cases.
- **Dispute Resolution**: Providing objective evidence in cases of agent performance disputes or customer complaint investigations.
- **Regulatory Compliance Auditing**: Demonstrating script adherence and compliance for regulatory audits in highly regulated industries.

3.6. SDG Goals

SDG 4: Quality Education

SPEAK-ALIGN serves as a means for continuous learning and skills improvement by providing accurate and helpful feedback regarding agent performance. The system further supports inclusive and equitable high-quality education by availing agents of individual coaching opportunities to improve their professional and communication skills.

SDG 8: Decent Work and Economic Growth

SPEAK-ALIGN promotes decent work by giving call center agents objective and unbiased performance appraisals and discarding bias-based performance evaluations. The system supports professional development through constructive feedback, which enables improved working conditions and career growth opportunities in the BPO sector.

SDG 9: Industry, Innovation, and Infrastructure

This project deploys the latest AI technologies, like speaker diarization, sophisticated speech recognition, and large language models, in order to innovate conventional quality assurance procedures. It enables a robust quality management infrastructure capable of growing with the industry and technology.

4. Hardware/Software Requirements

Hardware Requirements

Minimum Configuration:

- Processor: Intel Core i5 (8th Gen) or AMD Ryzen 5 or higher
- RAM: 16 GB DDR4
- Storage: 256 GB SSD with 100 GB free space for audio file processing
- Network: Stable internet connection with minimum 10 Mbps bandwidth for API calls

Recommended Configuration:

- Processor: Intel Core i7 (10th Gen) or AMD Ryzen 7 or higher
- RAM: 32 GB DDR4
- Storage: 512 GB NVMe SSD with 250 GB free space
- GPU: NVIDIA GPU with 8GB VRAM (optional, for faster processing)
- Network: High-speed internet connection with minimum 25 Mbps bandwidth

Software Requirements

Operating System:

- Windows 10/11 (64-bit) or Linux (Ubuntu 20.04 LTS or higher) or macOS 11 or higher

Programming Language and Framework:

- Python 3.10 or higher
- pip package manager

Required Libraries and Tools:

- Advanced speech recognition library for speech-to-text conversion
- Speaker diarization library (pyannote.audio or similar)
- Large language model API or local deployment for script analysis
- PyTorch or TensorFlow (backend for AI models)
- librosa or pydub (for audio file handling)

Additional Software:

- Text editor or IDE (VS Code, PyCharm)

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